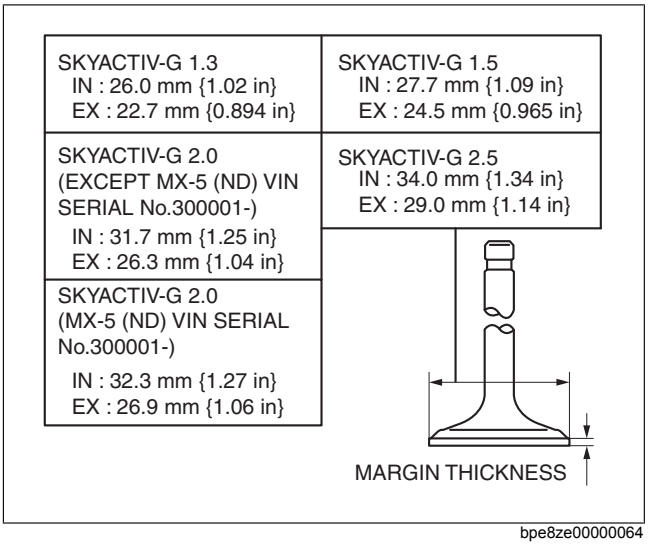
VALVE, VALVE GUIDE INSPECTION

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1. Measure the valve head margin thickness using a vernier caliper.
   - If it is less than the standard specification, replace the valve.

Standard valve head margin thickness
IN: 1.75—1.95 mm {0.0689—0.0767 in}
EX: 1.95—2.15 mm {0.0768—0.0846 in}



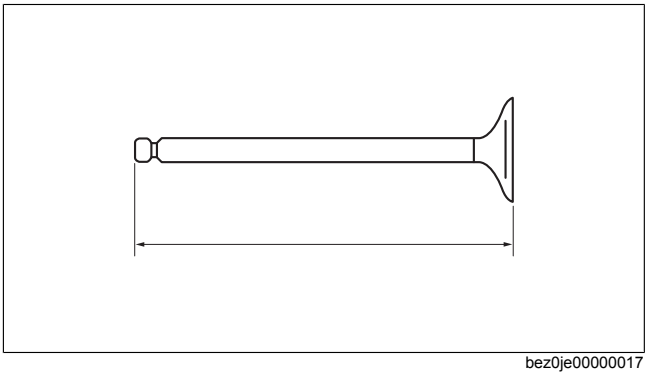
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2. Measure the length of the each valve using a vernier caliper.
   - If it is less than the minimum specification, replace the valve.

Standard valve length [SKYACTIV-G 1.3]
IN: 104.54—105.14 mm {4.1158—4.1393 in}
EX: 113.95—114.55 mm {4.4863—4.5098 in}

Standard valve length [SKYACTIV-G 1.5]
IN: 103.42—104.02 mm {4.0717—4.0952 in}
EX: 112.93—113.53 mm {4.4461—4.4696 in}

Standard valve length [SKYACTIV-G 2.0] Except MX-5 (ND) VIN Serial No.300001-
IN: 106.55—107.15 mm {4.1949—4.2185 in}
EX: 116.55—117.15 mm {4.5886—4.6122 in}



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Standard valve length [SKYACTIV-G 2.0] MX-5 (ND) VIN Serial No.300001-
IN:107.15-107.75 mm {4.2186-4.2421 in}
EX:117.15-117.75 mm {4.6123-4.6358 in}

Standard valve length [SKYACTIV-G 2.5]
IN: 107.00—107.60 mm {4.2127—4.2362 in}
EX: 117.09—117.69 mm {4.6099—4.6334 in}

Minimum valve length [SKYACTIV-G 1.3]
IN: 104.32 mm {4.1071 in}
EX: 113.73 mm {4.4776 in}

Minimum valve length [SKYACTIV-G 1.5]
IN: 103.20 mm {4.0630 in}
EX: 112.71 mm {4.4374 in}

Minimum valve length [SKYACTIV-G 2.0] Except MX-5 (ND) VIN Serial No.300001-
IN: 106.33 mm {4.1862 in}
EX: 116.33 mm {4.5799 in}

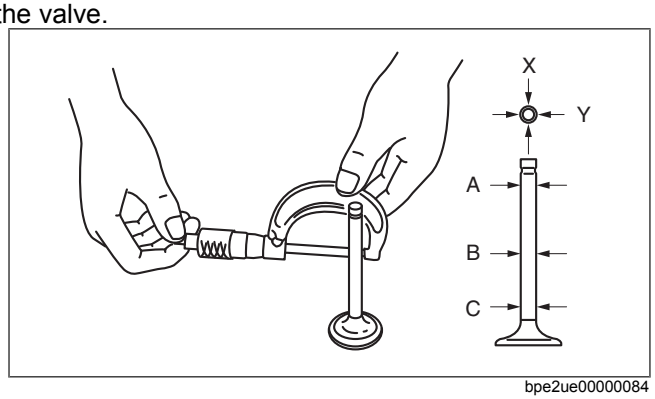
Minimum valve length [SKYACTIV-G 2.0] MX-5 (ND) VIN Serial No.300001-
IN: 106.33 mm {4.1862 in}
EX: 116.33 mm {4.5799 in}

Minimum valve length [SKYACTIV-G 2.5]
IN: 106.78 mm {4.2039 in}
EX: 116.87 mm {4.6012 in}.

3. Measure the valve stem diameter of each valve using the micrometer. Measurement positions total six and are in the X and Y directions, at three points (A, B, and C) as shown in the figure.
   - If it is less than the minimum specification, replace the valve.

Standard valve stem diameter
IN: 5.470—5.485 mm {0.2154—0.2159 in}
EX: 5.465—5.480 mm {0.2152—0.2157 in}

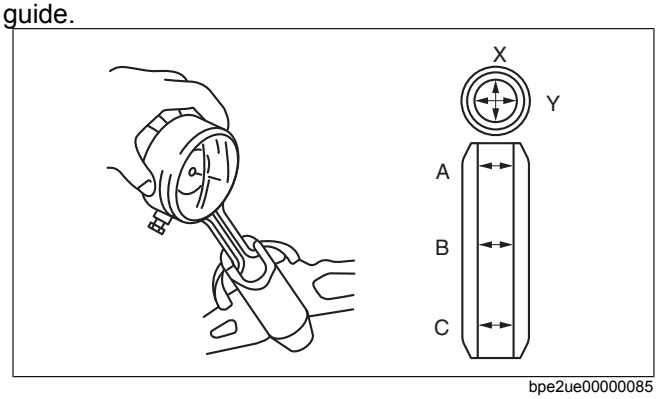
Minimum valve stem diameter
IN: 5.424 mm {0.2135 in}
EX: 5.419 mm {0.2133 in}



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4. Measure the inner diameter of each valve guide using the caliper gauge. Measurement positions total six and are in the X and Y directions, at three points (A, B, and C) as shown in the figure.
   - If it is not within the specification, replace the valve guide.

Standard valve guide inner diameter
IN: 5.510—5.530 mm {0.2170—0.2177 in}
EX: 5.510—5.530 mm {0.2170—0.2177 in}

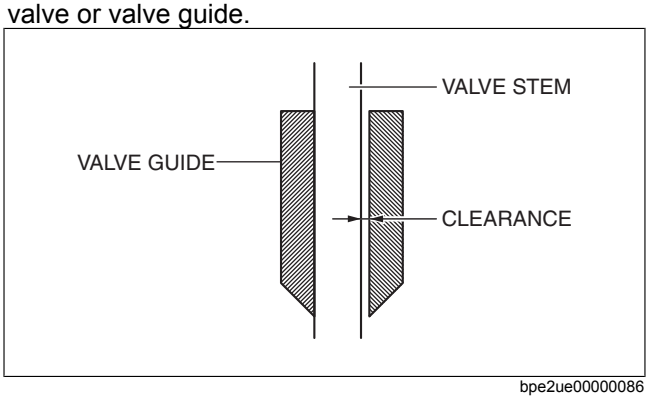


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5. Calculate the clearance between the valve stem and the valve guide by subtracting the inner diameter of the valve guide from the outer diameter of the corresponding valve stem.
   - If it exceeds the maximum specification, replace the valve or valve guide.

Standard clearance between valve stem and guide
IN: 0.025—0.060 mm {0.0010—0.0023 in}
EX: 0.030—0.065 mm {0.0012—0.0025 in}

Maximum clearance between valve stem and guide
0.10 mm {0.0039 in}

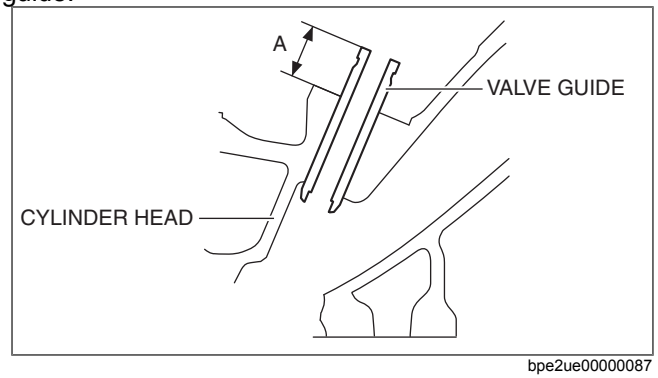


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6. Measure the projection height (dimension A) of each valve guide using the vernier caliper.
   - If it is not within the specification, replace the valve guide.

Standard valve guide projection height [SKYACTIV-G 1.3, SKYACTIV-G 1.5]
IN: 13.8—14.4 mm {0.544—0.566 in}
EX: 13.8—14.4 mm {0.544—0.566 in}

Standard valve guide projection height [SKYACTIV-G 2.0, SKYACTIV-G 2.5]
IN: 16.4—17.0 mm {0.646—0.669 in}
EX: 16.4—17.0 mm {0.646—0.669 in}



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